Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-5435
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

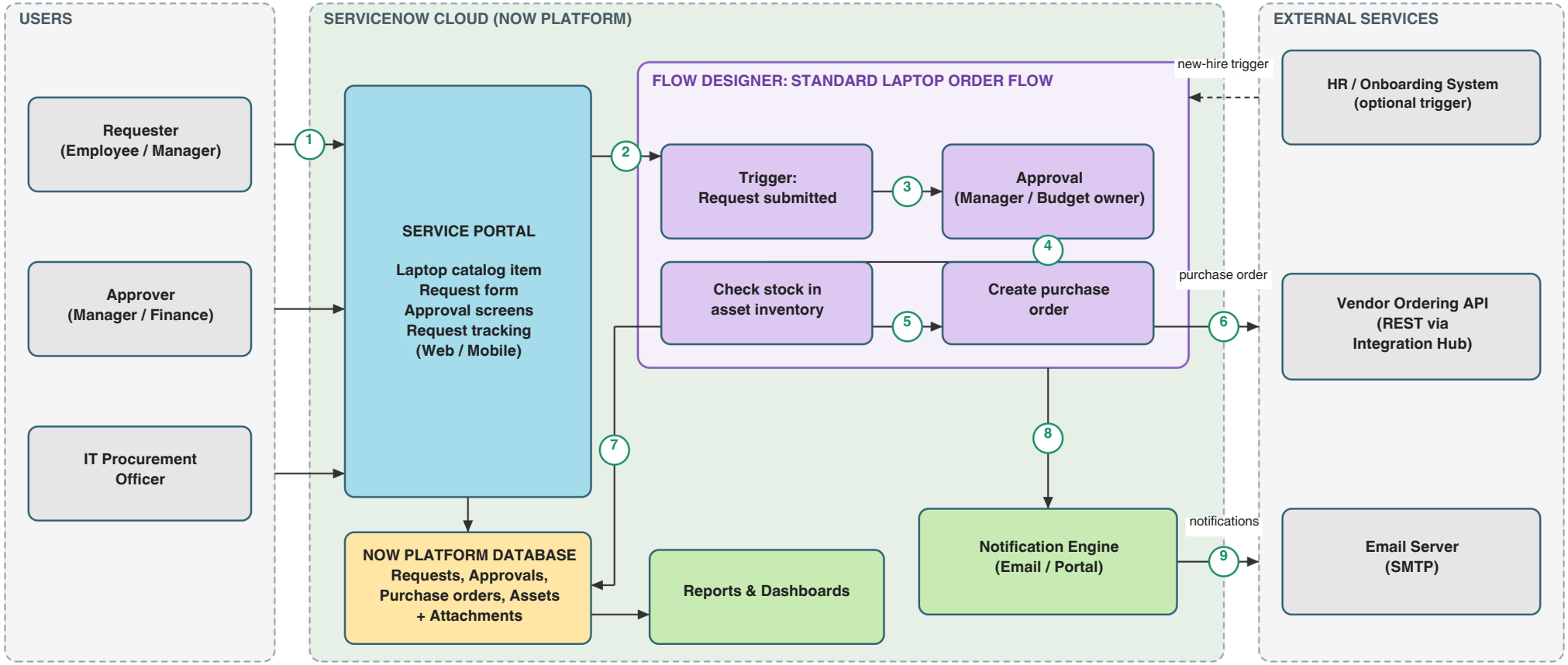
The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2.

Example: Automated standard laptop ordering on a cloud workflow platform (Flow Designer)

Data flow shown in the architecture diagram:

1.The requester submits the standard laptop catalog item on the Service Portal; approvers and IT staff also use the portal.
2.The portal creates the request record and triggers the Standard Laptop Order flow in Flow Designer.
3.The flow sends an approval request to the manager / budget approver and waits for the decision.
4.After approval, the flow checks stock in the asset inventory.
5.If no stock is available, the flow creates a purchase order automatically.
6.The purchase order is sent to the vendor through a REST API (Integration Hub).
7.Requests, approvals, purchase orders and asset records are stored in the Now Platform database.
8.The flow triggers notifications to the requester and approver at every status change.
9.Notifications are delivered through the email server; reports and dashboards read the stored data.

Architecture Diagram: Automating Standard Laptop Orders with Flow Designer



Legend: numbered circles follow the data flow steps on the previous page; blue = user interface, purple = application logic (Flow Designer), yellow = data storage, green = reporting / notifications, grey = users and external services; the dashed arrow shows an optional integration.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Requesters, approvers and IT staff interact through the Service Portal catalog item, request form and approval screens (web and mobile).	ServiceNow Service Portal (HTML, CSS, JavaScript, AngularJS); Catalog Item / Record Producer
2	Application Logic-1	Request submission: the flow is triggered when the laptop catalog item is submitted, and the entered details are validated.	Flow Designer (trigger), Catalog Client Scripts
3	Application Logic-2	Approval workflow: routes the request to the manager / budget approver, with reminders and escalation.	Flow Designer approval action, Approval Engine
4	Application Logic-3	Order processing: stock check, automatic purchase order creation, delivery and hand-over tasks, and automatic closure.	Flow Designer actions and subflows
5	Database	Stores requests, approvals, purchase orders, asset records and user data in relational tables.	Now Platform relational database
6	Cloud Database	Database service hosted on the cloud instance.	ServiceNow cloud instance database
7	File Storage	Attachments such as quotes, invoices and delivery documents linked to a request.	ServiceNow attachments (sys_attachment)
8	External API-1	Vendor ordering API used to place the purchase order and receive shipment details.	REST API through Integration Hub
9	External API-2	Email server used to send notifications to requesters and approvers.	Email notifications (SMTP)
10	Machine Learning Model	Not required in the current scope; can be added later, for example to forecast equipment demand.	Not applicable (optional: Predictive Intelligence)
11	Infrastructure (Server / Cloud)	Application deployment on cloud. **Local Server Configuration:** Not required; built and tested on the cloud instance. **Cloud Server Configuration:** ServiceNow instance with Flow Designer, Service Catalog and Integration Hub activated.	ServiceNow Cloud (Now Platform); Update Sets to move changes between instances

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source front-end frameworks used by the Service Portal.	AngularJS, Bootstrap
2	Security Implementations	Role-based access controls so users see only the data they need, approvals limited to assigned approvers, audit logging of changes, and encrypted communication.	Role-based ACLs, HTTPS / TLS, audit logs, SSO (optional)
3	Scalable Architecture	Multi-tier cloud architecture (interface, logic, data). Reusable flows and subflows allow new catalog items, such as monitors or accessories, to be added easily.	3-tier Now Platform architecture, Flow Designer subflows

S.No	Characteristics	Description	Technology
4	Availability	Hosted on a cloud platform managed by the provider, with error handling and retry of failed flow executions.	ServiceNow cloud hosting, Flow Designer error handling
5	Performance	Asynchronous flow execution, automatic approval of standard laptops within budget, and dashboards to monitor turnaround time.	Flow Designer, table indexing, Reports / Performance Analytics

References:

https://c4model.com/
https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
https://www.ibm.com/cloud/architecture
https://aws.amazon.com/architecture
https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d